

Q1 2021 (31 Aug Shift 2)

A free electron of 2.6 eV energy collides with H^+ ion. This results in the formation of a hydrogen atom in the first excited state and a photon is released. Find the frequency of the emitted photon ($h = 6.6 \times 10^{-34} \text{ Js}$)

(1) $1.45 \times 10^{16} \text{ MHz}$

(2) $0.19 \times 10^{15} \text{ MHz}$

(3) $1.45 \times 10^9 \text{ MHz}$

(4) $9.0 \times 10^{27} \text{ MHz}$

Q2 2021 (27 Aug Shift 2)

X different wavelengths may be observed in the spectrum from a hydrogen sample if the atoms are excited to states with principal quantum number $n = 6$? The value of X is

Q3 2021 (26 Aug Shift 1)

A particular hydrogen like ion emits radiation of frequency $2.92 \times 10^{15} \text{ Hz}$ when it makes transition from $n = 3$ to $n = 1$. The frequency in Hz of radiation emitted in transition from $n = 2$ to $n = 1$ will be :

(1) 0.44×10^{15}

(2) 6.57×10^{15}

(3) 4.38×10^{15}

(4) 2.46×10^{15}

Answer Key

Q1 (3)

Q2 (15)

Q3 (4)

Q1 (3)

For every large distance P.E. = 0

& total energy = $2.6 + 0 = 2.6\text{eV}$ Finally in first excited state of H atom total energy = -3.4eV Loss in total energy = $2.6 - (-3.4)$

$$= 6\text{eV}$$

It is emitted as photon

$$\lambda = \frac{1240}{6} = 206\text{ nm}$$

$$f = \frac{3 \times 10^8}{206 \times 10^{-9}} = 1.45 \times 10^{15}\text{ Hz}$$

$$= 1.45 \times 10^9\text{ Hz}$$

Q2 (15)

No. of different wavelengths = $\frac{n(n-1)}{2}$

$$= \frac{6 \times (6-1)}{2} = \frac{6 \times 5}{2} = 15$$

Q3 (4)

$$nf_1 = k \left(\frac{1}{1} - \frac{1}{3^2} \right)$$

$$nf_2 = k \left(1 - \frac{1}{2^2} \right)$$

$$\frac{f_1}{f_2} = \frac{8/9}{3/4} \Rightarrow f_2 = 2.46 \times 10^{15}$$

Option (4)